



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E CSE
Course Code & Title : AI3021 & IT in Agricultural System
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit I/Precision Farming/Remote Sensing
- **Course Outcome:** CO1
- **Topic Learning Outcome:** 1a
- **Activity Chosen:** Flipped Classroom
- **Justification:**
 - Remote sensing involves complex concepts such as electromagnetic spectrum analysis, sensor technologies, and satellite data interpretation. These topics benefit significantly from visual aids, interactive discussions, and real-world case studies. Since students often have prior exposure to basic physics and data analysis, a flipped classroom allows them to review foundational content independently through videos, articles, and satellite imagery before class.
 - During class, they engage in deeper discussions focusing on applications like flood mapping, crop monitoring, and urban planning, collaborate in teams to solve case-based problems using remote sensing techniques, and receive targeted feedback from faculty to clarify misconceptions and reinforce learning. This method promotes active learning, improves student preparedness, and bridges theory with practical application—making it ideal for a multidisciplinary subject like Remote Sensing.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - To make the learning experience more engaging, a **flipped classroom activity** was organized for **Remote Sensing** on **4th August 2025**. The learning materials, which included the fundamentals of remote sensing and sample satellite imagery, were uploaded to the **Canvas platform** on **31st July 2025**, three days before the class. Students were asked to go through the materials independently and take short notes on key topics such as the types of sensors (passive and active), platforms (satellites and drones), and real-world applications like agriculture, disaster management, and urban planning.
 - On the day of the activity, the class was divided into **17 teams**, each consisting of three students. Every team spent about 10 minutes discussing the concepts they had studied. Following this, each group received a short case study examples included flood detection, crop health monitoring, and urban expansion analysis and they were asked to identify the appropriate remote sensing techniques and justify their choices based on the pre-class materials.

- During the presentations, several teams shared thoughtful insights. **Ms. Asrar Siddiq** presented her team's findings clearly, using diagrams and satellite examples to explain the concepts. The peer-sharing session encouraged collaborative learning and critical thinking. The faculty facilitated the discussion, offered feedback, clarified misconceptions, and connected the ideas to practical, real-world applications.
- Overall, the flipped classroom implementation helped students become more active learners, enhanced their preparedness, and promoted a deeper understanding of remote sensing technologies through teamwork and guided exploration.

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO8	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO5	PO5	PO10	PO12	PSO2
	3	3	3	1	2	2	2
Justification for correlation	Student apply the basic knowledge of environmental sciences and IT technologies to find easy solutions of some difficult agricultural methods..	Student identify and analyse the principles of remote sensing applications.	Student able to deploy the basic concepts of GPS and mapping software.	Student use research-based knowledge tools such as Power BI and Tableau including data visualization and manipulate data to provide valid conclusion.	Student interpret data visualization and prepare as a report or presentation	Student exhibit To explore emerging applications for Precision agriculture.	Students can apply IoT and cloud-based geospatial data processing to design reliable precision agriculture solutions.

- **Images / Screenshot of the practice:**



Fig 1:Asrar Presented the ideas

- **Reflective Critique:**

- **Self-paced Learning Experience:**

Students appreciated the flexibility that the flipped classroom provided. They mentioned that being able to learn at their own pace and review materials multiple times before class helped them understand the concepts more clearly. It also allowed them to manage their study schedules according to their convenience.

- **Enhanced Conceptual Understanding:**

Many students expressed that studying the basics through pre-class videos and reading materials made them more confident during in-class discussions. They felt better prepared to participate actively and to apply what they had learned to real-world scenarios.

- **Exploration and Curiosity:**

Students shared that this approach encouraged them to explore additional resources beyond the given materials. They were motivated to learn about new satellite technologies, image processing techniques, and recent remote sensing applications in environmental studies.

- **Collaborative Learning:**

Working in small teams helped students share ideas and clarify doubts collectively. They enjoyed peer discussions and found that explaining concepts to their teammates improved their own understanding. It also helped in building communication and teamwork skills.

- **Faculty Guidance and Support:**

Students mentioned that they benefited greatly from the faculty's guidance during class activities. Immediate feedback on their case study presentations and clarifications of misconceptions strengthened their learning outcomes.

- **Active Classroom Engagement:**

The flipped format transformed the classroom into a more dynamic learning environment. Instead of passive listening, students were actively engaged in analyzing data, interpreting satellite imagery, and connecting theoretical concepts with real-world applications.

- **Stakeholder Observation:**

Faculty members observed that students came to class with better preparedness and showed higher levels of curiosity and participation. The approach also encouraged critical thinking and independent learning, aligning well with outcome-based education principles.

- **Overall Learning Impact:**

Both students and faculty agreed that the flipped classroom method was effective in bridging theory and practice. It made the learning process more student-centered, engaging, and reflective, leading to improved understanding and long-term retention of remote sensing concepts.

- **Benefit of the practice:**

- The flipped classroom model allows the teacher to spend more individualized time with students during class. Since basic concepts are learned beforehand, class hours can be dedicated to addressing specific doubts, offering personalized guidance, and supporting students based on their unique learning needs.
- Students work together in small groups to apply theoretical knowledge to real-world case studies. This setup promotes teamwork, peer learning, and communication skills, as they discuss, debate, and solve problems collaboratively.
- Since core learning materials are available online, students who miss class due to illness or personal reasons can easily catch up by reviewing the shared resources. This ensures continuity in learning without gaps.

- **Challenges faced in implementation:**

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://en.wikipedia.org/wiki/Flipped_classroom
- <https://studv.com/teach/flippedclassroom.html?msocid=0e0d746675f569a42f72622174ae6862>
- <https://www.extramarks.com/blogs/teachers/flipped-classrooms/>
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Signature of Faculty Member

HOD